Indian Stock Market Trend Analysis Using Time-Series Forecasting CRISP-DM

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***Abstract*—This project benchmarks univariate forecasting for the NIFTY 50 OHLCV variables using three classical methods: Naive, SeasonalNaive (lag=5 trading days), and ARIMA. Performance is evaluated using RMSE and MAPE under two protocols: (A) fixed-horizon walk-forward forecasting for $H \in \{1, 7, 30\}$ and (B) full test-set forecasting without updating. Results show that strong persistence/seasonal baselines remain highly competitive for index prices and that evaluation protocol design materially affects both error magnitude and model ranking. Residual diagnostics are reported for a representative case to assess remaining autocorrelation.**

***Index Terms*—Stock Market Prediction, Machine Learning, CRISP-DM, NIFTY 50, Time-Series Forecasting**

I. Introduction

The Indian stock market is one of the fastest-growing financial markets in the world, playing a crucial role in the country's economic development. It operates through two major exchanges: the Bombay Stock Exchange (BSE) and the National Stock Exchange (NSE), both of which facilitate trading in equities, derivatives, commodities and debt instruments. With millions of investors, ranging from retail traders to institutional investors, the Indian stock market significantly impacts economic policies, business growth and investment strategies. Over the past few decades, advancements in technology and regulatory reforms have enhanced market efficiency, making it a preferred destination for both domestic and international investors [1].

The sheer volume of trade conducted in the Indian stock market underscores its importance. The NSE alone records an average daily trading volume exceeding $20 billion [1], making it one of the most active markets globally. This high liquidity ensures efficient price discovery and minimal market manipulation. The market's deep integration with global financial systems means that any significant movement in Indian stocks often reflects broader economic trends. Additionally, the participation of Foreign Institutional Investors (FIIs) further amplifies the market's dynamism. Understanding and predicting stock market trends, therefore, becomes essential not only for investors but also for policymakers and economists who rely on accurate market insights for decision-making.

Studying the Indian stock market is critical due to its influence on national and international economic stability. Unlike many developed markets, the Indian stock market is highly volatile, driven by factors such as political changes, monetary policies, global trade relations and investor sentiment. Predicting market movements accurately can mitigate financial risks, optimize investment strategies and enhance market stability. Traditional forecasting methods often fail to capture the complexities of the stock market, necessitating the use of advanced analytical techniques, such as Machine Learning (ML) [2], to improve prediction accuracy. The motivation for selecting the Indian stock market lies in its high growth potential, rapid digitization and increasing retail participation, making it a fertile ground for applying modern predictive models.

Within the Indian stock market, the NIFTY 50 [3] index serves as a crucial benchmark, representing the top 50 companies listed on the NSE. This index is widely tracked by analysts, fund managers and investors due to its broad sectoral representation and role as an indicator of market performance. The selection of NIFTY 50 for this study is justified by its high trading volume, strong market capitalization and ability to reflect overall economic conditions. Unlike individual stock analysis, which is often influenced by company-specific factors, studying an index like NIFTY 50 provides a more comprehensive understanding of market trends. Additionally, the NIFTY 50 index is less susceptible to manipulation and speculative fluctuations, making it a reliable dataset for predictive modeling.

One widely used classical approach for financial time-series forecasting is the Auto-Regressive Integrated Moving Average (ARIMA) family [4]. In this project, we benchmark univariate forecasting for NIFTY 50 OHLCV variables by comparing ARIMA to two strong baselines (Naive and SeasonalNaive with lag=5 trading days). Performance is reported under two evaluation protocols: (A) fixed-horizon walk-forward forecasting for $H \in \{1, 7, 30\}$ and (B) full test-set forecasting without updating. The goal is not to "beat the market," but to produce a transparent, reproducible evaluation of what can be achieved using historical values alone.

To structure this approach, the study follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology [5]. The process includes business understanding, data collection, preprocessing, modeling, evaluation and deployment, ensuring a systematic approach to stock market trend analysis. The dataset consists of historical NIFTY 50 stock prices obtained from the *yfinance* API, incorporating key financial indicators such as opening and closing prices, trading volume [6].

Interpretability is addressed through model transparency (explicit ARIMA orders per feature) and forecast diagnostics (error metrics and residual tests). Residual analysis and Ljung–Box testing are used to assess whether remaining autocorrelation persists after modeling, which helps explain where simple baselines are sufficient and where ARIMA provides measurable benefit.

II. Aim, Objectives and Expected Outcomes

Aim: The aim of this project is to benchmark reproducible time-series forecasting for the NIFTY 50 OHLCV variables using transparent, classical models and strong persistence baselines.

Objectives:

- Acquire and describe the OHLCV dataset and its time span, then construct a leakage-free train/test split.
- Implement three univariate forecasting methods: Naive, SeasonalNaive (lag=5 trading days), and ARIMA.
- Select one ARIMA order per feature using AIC on the training split and reuse it consistently across all evaluations.
- Evaluate performance using RMSE and MAPE under two protocols: fixed-horizon walk-forward (Protocol A) and full test-set no-updating (Protocol B).
- Provide diagnostic evidence via residual analysis for a representative case (Close, 1-day ahead, Protocol A).

Expected Outcomes:

- A fully reproducible notebook and exported artifacts (tables, figures, and an HTML dashboard) packaged with the submission.
- A clear comparison showing when simple baselines are sufficient and when ARIMA provides measurable benefit.

III. Research Questions

- **RQ1:** How do Naive, SeasonalNaive(lag=5), and ARIMA compare on OHLCV forecasting accuracy under Protocol A (walk-forward) and Protocol B (no updating)?
- **RQ2:** How does forecast horizon ($H \in \{1, 7, 30\}$) affect error growth under Protocol A?
- **RQ3:** Do residual diagnostics for a representative case (Close, 1-day ahead, Protocol A) indicate remaining autocorrelation?

These questions emphasize reproducible benchmarking and the impact of evaluation protocol design on observed model rankings and focus on reproducible benchmarking of univariate baselines and ARIMA under two evaluation protocols, including the effect of horizon length and residual diagnostics for a representative case.

IV. Methodology

This study follows the CRISP-DM framework, ensuring a structured and systematic approach to stock market trend prediction. The methodology consists of six key phases: business understanding, data understanding, data preparation, modeling, evaluation and deployment [5].

1. Business Understanding: This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) [5] framework, ensuring a structured and systematic approach to stock market trend prediction. The methodology includes six key phases, from business understanding to deployment, ensuring a comprehensive analysis of stock market movements.

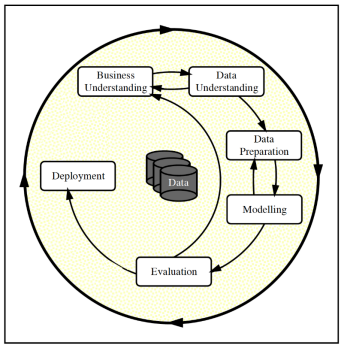


Fig. 1. Phases of the CRISP-DM reference model [7]

2. Data Understanding: We collect daily NIFTY 50 OHLCV data using the *yfinance* API [6]. After retrieval and cleaning, the dataset contains 1975 observations with five target variables (Open, High, Low, Close, Volume) spanning 2018-01-02 to 2026-01-05. The split is time-ordered (80% train / 20% test), yielding 1580 training points and 395 test points. Because financial series are non-stationary and evaluation is sensitive to leakage, all modeling uses only past information relative to the forecast origin.

3. Data Preparation: Data preprocessing is a critical step in ensuring the accuracy and reliability of stock market trend forecasting. To ensure the accuracy and reliability of the predictive models, the collected data from yfinance API undergoes data preprocessing :

- Sort the series chronologically and validate the date range.
- Handle missing values using forward/back fill (to preserve time continuity).
- Use a time-ordered split (80% train, 20% test) to avoid leakage.

4. Modeling: This work models each target variable (Open, High, Low, Close, Volume) as a *separate univariate time series* to keep the learning problem tractable and reproducible under the project constraints. We compare three baselines/models:

- **Naive**: $\hat{y}_{t+h} = y_t$ (random-walk baseline).

- **Seasonal Naive (lag=5)**: $\hat{y}_{t+h} = y_{t-5}$ to reflect a weekly trading pattern (5 trading days).
- **ARIMA**(p, d, q): a classical linear time-series model that combines autoregression and moving-average terms, with optional differencing to stabilize the mean. [4]

Parameter tuning. For each feature, we select a single ARIMA order (p, d, q) on the training split using an AIC-based search (reported in Table I). These per-feature orders are then reused consistently across evaluation protocols and horizons to avoid overfitting-by-selection and to keep the experiment reproducible.

TABLE I
SELECTED ARIMA ORDERS PER FEATURE (AIC-BASED SELECTION ON THE TRAINING SPLIT).

Feature	ARIMA(p,d,q)	AIC
Open	(2,0,2)	20413.62
High	(2,0,2)	19741.54
Low	(1,0,2)	20349.06
Close	(1,0,0)	20350.92
Volume	(2,0,2)	41332.45

Table I reports one AIC-selected ARIMA order per feature fitted on the training split. Most price series (Open/High/Low/Close) are modeled with $d = 0$ under the constrained search used here, while Volume selects a higher-AIC model family due to its different scale and dynamics. Importantly, these ARIMA orders are reused across horizons and protocols; the horizons are part of the evaluation design (Protocol A), not separate per-horizon model selection.

Each model is fitted on the training split and evaluated on the held-out test split using time-ordered forecasting. Protocol A uses walk-forward evaluation at fixed horizons ($H \in \{1, 7, 30\}$) where history is updated as new observations arrive, while Protocol B produces a full test-set forecast without updating after training. Residuals are computed as $e_t = y_t - \hat{y}_t$; diagnostic plots and Ljung–Box testing are reported for a representative case (Close, 1-day ahead, Protocol A).

5. Evaluation: We evaluate Naive, SeasonalNaive(lag=5), and ARIMA using RMSE and MAPE under two protocols. Protocol A is a fixed-horizon walk-forward evaluation for $H \in \{1, 7, 30\}$ in which the history is updated as new observations become available. Protocol B is a full test-set forecast without updating, which stresses error accumulation and robustness. For Volume, MAPE is not interpreted due to instability when denominators are close to zero, Volume conclusions are based on RMSE.

6. Deployment: All results are exported as reproducible artifacts (CSV tables and figures) and an HTML dashboard (`dashboard.html`) that allows inspection of predictions and residual behavior.

7. Interpretability: Interpretability is provided through explicit model specification (reported ARIMA orders) and diagnostic analysis (residual time-series, ACF/PACF, and Ljung–Box testing) to assess whether residual autocorrelation remains.

V. RESULTS & ANALYSIS

This section presents the outcomes of the ARIMA-based forecasting framework applied to NIFTY 50 stock market indicators. Two types of results were generated and analyzed:

- Fixed-horizon Forecasting.
- Full Test Set Forecasting.

A. Fixed-horizon Forecasting:

Protocol A performs walk-forward evaluation at fixed horizons $H \in \{1, 7, 30\}$. At each time step, the model forecasts H days ahead using only past information, then the true next observation is revealed and appended before moving forward. This mirrors a practical setting where forecasts are continually produced as new market data arrives.

TABLE II
PROTOCOL A (FIXED-HORIZON WALK-FORWARD): FORECASTING ERROR PER FEATURE AND HORIZON. RMSE IS IN RAW UNITS (INDEX POINTS FOR OHLC; TRADED UNITS FOR VOLUME). MAPE IS REPORTED IN PERCENT; FOR VOLUME, MAPE IS OMITTED (–) DUE TO NUMERICAL INSTABILITY WHEN DENOMINATORS APPROACH ZERO.

Feature	H	RMSE Naive	RMSE Seasonal	RMSE ARIMA	MAPE Naive	MAPE Seasonal	MAPE ARIMA
Open	1	206.5591	477.2015	209.8468	0.6091	1.5323	0.6246
Open	7	514.9978	620.5728	530.1113	1.6667	2.0694	1.7410
Open	30	920.3031	1006.2229	938.1170	2.9953	3.3141	3.0759
High	1	162.9183	447.9460	167.8950	0.4755	1.4380	0.4921
High	7	483.1633	599.1912	498.2139	1.5689	2.0131	1.6330
High	30	906.2748	1000.6737	929.9132	2.9401	3.3160	3.0322
Low	1	187.6004	465.7902	192.4771	0.5498	1.4803	0.5666
Low	7	512.4173	618.2272	528.2249	1.6791	2.1027	1.7503
Low	30	922.4189	1014.7176	940.7748	3.0270	3.3353	3.1046
Close	1	181.0681	458.5799	184.7490	0.5460	1.4669	0.5592
Close	7	501.0616	612.3951	514.1635	1.6416	2.0796	1.7057
Close	30	913.0757	1004.1081	930.3018	2.9693	3.3288	3.0352
Volume	1	124323.2579	136274.8462	100095.7154	–	–	–
Volume	7	141525.7397	141362.2573	105128.8981	–	–	–
Volume	30	138387.3098	137270.7216	101965.8213	–	–	–

1) Observations & Analysis: Table II shows the expected *horizon effect*: for the OHLC series, both RMSE and MAPE generally increase as H grows from 1 to 30 days, reflecting greater uncertainty and error accumulation further into the future. Under Protocol A, the **Naive** baseline is consistently the strongest performer for OHLC (e.g., Close: RMSE 181.07 at 1-day vs. 913.08 at 30-day), indicating that over this dataset the immediate past value is a difficult-to-beat predictor.

For **Volume**, ARIMA achieves the lowest RMSE across horizons, but we do not interpret MAPE due to numerical instability of percentage errors when denominators approach zero. Overall, Protocol A produces substantially lower errors than Protocol B (Table III), consistent with the benefit of walk-forward updating when new observations are continually incorporated.

2) Key Findings:

- **Errors increase with horizon.** Across OHLC, RMSE/MAPE rise from 1-day to 30-day, which is consistent with increasing forecast uncertainty over longer horizons.
- **Naive dominates OHLC under Protocol A.** For Open/High/Low/Close, Naive yields the lowest RMSE at all horizons, while ARIMA is close but not consistently superior.
- **Volume behaves differently.** ARIMA provides the lowest RMSE for Volume across horizons, suggesting exploitable autocorrelation in the traded-volume process under this setup.

- **Protocol choice matters.** Walk-forward updating (Protocol A) yields materially smaller errors than no-update forecasting (Protocol B), highlighting distribution shift and error accumulation effects in longer deployments.

3) Potential Reasons for These Results: Two factors plausibly explain the pattern. First, equity index prices often exhibit strong persistence (random-walk-like behavior), making last-value baselines surprisingly competitive for short-range prediction. Second, the modeling here is *univariate*: macroeconomic/news drivers and cross-feature interactions are not represented, limiting ARIMA's ability to outperform a persistence baseline on OHLC. In contrast, Volume can exhibit stronger short-memory structure (e.g., weekly trading rhythm), which can be captured by AR/MA dynamics and can explain ARIMA's RMSE advantage for that target.

B. Full Test Set Forecasting

Protocol B trains once on the training split and forecasts the entire test interval without updating. This setting is intentionally stricter than Protocol A and highlights multi-step drift and distribution shift effects. We therefore expect larger errors and potentially different model rankings than in walk-forward evaluation.

TABLE III
PROTOCOL B (FULL TEST-SET FORECAST, NO UPDATING): MODEL COMPARISON PER FEATURE. RMSE IS IN RAW UNITS. MAPE FOR VOLUME IS UNSTABLE AND NOT INTERPRETED.

Feature	Model	RMSE	MAPE (%)
Open	Naive	1686.9759	5.8807
Open	SeasonalNaive(lag=5)	1950.2570	6.8693
Open	ARIMA(2,0,2)	2162.7088	7.5049
High	Naive	1766.2090	6.1760
High	SeasonalNaive(lag=5)	1987.7618	7.0215
High	ARIMA(2,0,2)	2212.7566	7.7503
Low	Naive	3307.1118	12.7615
Low	SeasonalNaive(lag=5)	2359.2207	8.3997
Low	ARIMA(1,0,2)	3835.7780	14.8613
Close	Naive	2837.4959	10.7113
Close	SeasonalNaive(lag=5)	2246.4937	8.0265
Close	ARIMA(1,0,0)	2967.4698	11.2176
Volume	Naive	709251.1298	–
Volume	SeasonalNaive(lag=5)	370004.5227	–
Volume	ARIMA(2,0,2)	145306.6387	–

1) Observations & Analysis: Under Protocol B (no updating), errors increase sharply for all targets compared to Protocol A, which is consistent with distribution shift over time and recursive forecast drift. Model ranking is feature-dependent: Naive yields the lowest RMSE for Open and High, SeasonalNaive(lag=5) is best for Low and Close, and ARIMA achieves the lowest RMSE for Volume (Table III). These outcomes indicate that when the model cannot refresh on new data, simple persistence/seasonal structure can be more robust than a fixed ARIMA model fitted only once. Because Volume yields unstable MAPE values, the MAPE panels should be interpreted with caution, Table results are the primary source for OHLC interpretation.

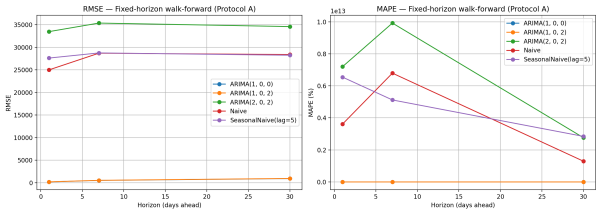


Fig. 2. Protocol A: RMSE and MAPE aggregated across features for fixed horizons (walk-forward updating).

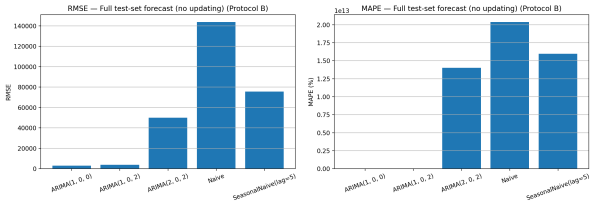


Fig. 3. Protocol B: RMSE and MAPE aggregated across features for full test-set forecasting (no updating).

2) Key Findings: Protocol A: Naive is consistently best for OHLC, RMSE increases with horizon (expected).

Protocol B: SeasonalNaive improves Close and Low, Naive remains best for Open/High, ARIMA is best for Volume RMSE.

3) Potential Reasons for These Results: The performance differences observed across features can be explained by two dominant effects. First, OHLC index prices exhibit strong persistence, so the Naive baseline (last value) is difficult to beat, particularly under walk-forward evaluation (Protocol A). Second, under no-updating evaluation (Protocol B), short weekly recurrence can become more visible in Close and Low, which explains why SeasonalNaive(lag=5) improves these features relative to Naive and ARIMA. Volume behaves differently in scale and dynamics; in this study ARIMA achieves the best Volume RMSE under Protocol B, while MAPE is not interpreted for Volume due to instability when denominators approach zero. Finally, all models are univariate, so external drivers and cross-feature relationships are intentionally excluded.

C. Evaluation Protocols

We report results using two complementary evaluation protocols to separate *short-range operational forecasting* from *long-range, no-update deployment* behavior.

Protocol A (fixed-horizon walk-forward). For horizons $H \in \{1, 7, 30\}$ days ahead, we perform walk-forward evaluation over the test interval: at each step t, the forecast is produced using only information available up to t, then the history is updated with the newly revealed ground-truth before moving to $t+1$. This protocol approximates a realistic setting where the model is periodically refreshed as new market data arrives.

Protocol B (full test-set, no updating). The model is trained once on the training split and then forecasts the entire

test interval without any updating. This setting stresses robustness under distribution shift and highlights error accumulation in longer, recursive use.

Across both protocols we evaluate **Naive**, **Seasonal-Naive(lag=5)**, and **ARIMA** using RMSE and MAPE; however, **MAPE for Volume is not interpreted** because near-zero denominators can produce numerically unstable percentage errors.

D. Residual Diagnostics

We assess adequacy of the ARIMA residuals for Close (1-day ahead, Protocol A) using both visual diagnostics (residual time-series; ACF/PACF) and a Ljung–Box test. The ACF/PACF plots can show mild short-lag structure, but the Ljung–Box p-values at lags 10/20/30 (Table IV) do not reject the null of no autocorrelation at the 5% level. Overall, these diagnostics suggest that the remaining dependence is limited under this configuration, while acknowledging that market non-stationarity and omitted exogenous factors can still contribute to forecast error.

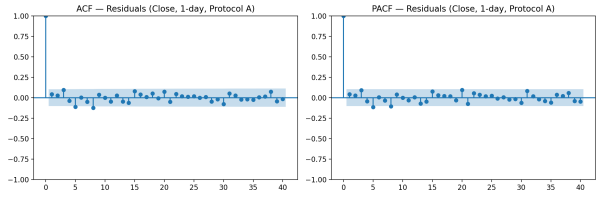


Fig. 4. Residual ACF/PACF for Close (1-day ahead), Protocol A.

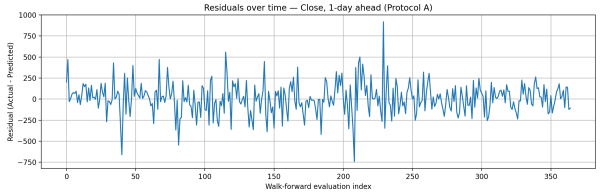


Fig. 5. Residuals over time for Close (1-day ahead), Protocol A.

TABLE IV
LJUNG–BOX TEST FOR CLOSE RESIDUALS (PROTOCOL A, 1-DAY AHEAD).

Lag	LB stat	p-value
10	16.6427	0.0827
20	26.2151	0.1588
30	31.8091	0.3765

The Ljung–Box test (lags 10/20/30) yields p-values 0.083/0.159/0.376, indicating no statistically significant residual autocorrelation at the 5% level.

1) Observations & Analysis: Residuals $e_t = y_t - \hat{y}_t$ are centered around zero with intermittent bursts during volatile periods, which is consistent with time-varying variance in financial series. While the ACF/PACF may show small early-lag spikes, the Ljung–Box results (p-values 0.083/0.159/0.376 at lags 10/20/30) indicate no statistically significant residual autocorrelation at the 5% level for this diagnostic case. This supports the interpretation that the remaining error is driven more by regime changes and volatility clustering than by strong unmodeled linear autocorrelation.

2) Key Findings:

- Diagnostics are reported for **Close, 1-day ahead, Protocol A**. Ljung–Box p-values at lags 10/20/30 do not reject the null of no autocorrelation at the 5% level, indicating no strong remaining linear dependence in this representative case.
- Residual inspection supports the interpretation that performance differences in Tables II III are primarily driven by protocol design (updating vs. no updating) and horizon length rather than by a large unmodeled autocorrelation structure.

3) Deployment - Web application: To support practical deployment and result inspection, we provide a lightweight web-based dashboard exported as an interactive HTML application. The dashboard enables feature-wise exploration (Open/High/Low/Close/Volume) under Protocol B via a dropdown selector and overlays Actual vs Forecast for ARIMA, Naive, and SeasonalNaive on the full test interval. In addition, it visualizes the ARIMA residual distribution as a histogram to help assess systematic bias and error spread. This deliverable (included as outputs/dashboard.html) provides a reproducible, user-facing interface for validating model behavior beyond summary tables and static plots.



Fig. 6. Forecast dashboard example (Feature: Open, Protocol B). Top: Actual vs. Forecast over the full test set. Bottom: ARIMA residual histogram.

VI. DISCUSSION

This section discusses the results by directly answering the research questions. The discussion is grounded in the two evaluation protocols (Protocol A: fixed-horizon walk-forward; Protocol B: full test-set forecasting without updating) and the reported metrics/tables.

A. RQ1: How do Naive, SeasonalNaive(lag=5), and ARIMA compare under Protocol A and Protocol B?

Under Protocol A (walk-forward), Naive consistently achieves the lowest error for the OHLC price series across all horizons $H \in \{1, 7, 30\}$ (Table II). ARIMA is generally close but does not provide consistent improvements over persistence, while SeasonalNaive(lag=5) performs worse than Naive for OHLC in this protocol. This indicates that, for the evaluated period and univariate setting, short-memory persistence dominates model ranking when the history is continually updated with new observations.

Under Protocol B (no updating), errors increase substantially for all targets relative to Protocol A (Table III), which is expected due to multi-step drift and distribution shift effects. In this stricter setting, the ranking becomes feature-dependent: SeasonalNaive(lag=5) performs best for Low and Close, while Naive remains best for Open and High. For Volume, ARIMA achieves the lowest RMSE among the evaluated methods. Overall, the results show that evaluation protocol design is a first-order factor: updating vs. no updating materially affects both the magnitude of error and the relative competitiveness of models.

B. RQ2: How does horizon length affect error growth under Protocol A?

Across OHLC under Protocol A, both RMSE and MAPE generally increase as the horizon grows from 1 to 30 days (Table II). This pattern is expected in financial time series because uncertainty compounds as forecasts extend further into the future. The horizon effect is practically important: performance that appears strong at 1-day ahead can degrade markedly at 7-day and 30-day horizons, even when using the same model family and the same updating procedure.

For Volume, RMSE varies by horizon and ARIMA provides the lowest RMSE across horizons in Protocol A. However, percentage-based errors (MAPE) are not interpreted for Volume because near-zero denominators can yield numerically unstable values; Volume conclusions are therefore based on RMSE rather than MAPE.

C. RQ3: Do residual diagnostics for a representative case indicate remaining autocorrelation?

To assess model adequacy beyond aggregate error metrics, we analyze residuals for a representative case (Close, 1-day ahead, Protocol A) using residual time-series inspection, ACF/PACF plots, and the Ljung–Box test (Table IV and Figures 4–5). The ACF/PACF plots may suggest mild short-lag structure, but the Ljung–Box p-values at lags 10/20/30 (0.083/0.159/0.376) do not reject the null hypothesis of no autocorrelation at the 5% level. Therefore, the diagnostics do not provide evidence of strong remaining linear dependence in this configuration. This supports the interpretation that the dominant performance differences observed across protocols and horizons are more plausibly explained by updating vs. no-updating behavior, horizon length, and non-stationary market dynamics rather than by a large unmodeled autocorrelation structure.

VII. Limitations

This study is restricted to **univariate** forecasting of OHLCV and therefore excludes exogenous drivers (news, macro signals) and cross-feature interactions that can influence market movements. Financial series are non-stationary and can exhibit regime shifts and volatility clustering; consequently, results depend on the chosen time period and the fixed train/test split. Finally, percentage-based errors (MAPE) are numerically unstable for **Volume** when denominators approach zero; for Volume we therefore interpret only RMSE and treat MAPE as not applicable.

VIII. Future Work

Future work could incorporate exogenous variables (e.g., macroeconomic indicators) and multivariate models to test whether additional information improves over strong persistence/seasonal baselines. A broader diagnostic suite could also be generated for all features and horizons.

IX. Conclusion

This project applied the CRISP-DM methodology to a real-world forecasting task using daily NIFTY 50 OHLCV data. We benchmarked three transparent univariate methods—Naive, SeasonalNaive(lag=5 trading days), and ARIMA—under two complementary evaluation protocols: (A) fixed-horizon walk-forward forecasting for $H \in \{1, 7, 30\}$ and (B) full test-set forecasting without updating. Performance was measured using RMSE and MAPE, with the exception that MAPE was not interpreted for Volume due to numerical instability when denominators approach zero.

The main findings are as follows. Under Protocol A, Naive consistently achieved the lowest error for the OHLC price series across all horizons, while ARIMA remained close but did not provide consistent improvements over persistence; errors increased with horizon length as expected. Under Protocol B, errors were substantially larger and model rankings became feature-dependent: SeasonalNaive(lag=5) performed best for Low and Close, Naive remained best for Open and High, and ARIMA achieved the lowest RMSE for Volume. Residual diagnostics for a representative case (Close, 1-day ahead, Protocol A) did not show strong evidence of remaining autocorrelation at the 5% level based on Ljung–Box testing.

Overall, the results indicate that strong persistence and simple seasonal baselines are difficult to outperform for univariate index-price forecasting, and that evaluation protocol design (updating vs. no updating) materially influences both absolute error and relative model ranking. The delivered notebook and exported artifacts provide a reproducible baseline for future extensions that incorporate exogenous information or multivariate modeling.

References

[1] "An Introduction to Indian Stock Markets." [Online]. Available: https://www.jmfinancialservices.in

[2] A. Mellouk and A. Chebira, *Machine Learning*. BoD – Books on Demand, Jan. 2009, google-Books-ID: RTqhDwAAQBAJ.

[3] "NIFTY 50," Feb. 2025, page Version ID: 1275437589. [Online]. Available: https://en.wikipedia.org/w/index.php?title=NIFTY$_50oldid$ = 1275437589

[4] "Autoregressive Integrated Moving Average (ARIMA) Prediction Model." [Online]. Available: https://www.investopedia.com/terms/a/autoregressive-integrated-moving-average-arima.asp

[5] "Cross-industry standard process for data mining," Aug. 2024, page Version ID: 1242296671. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Cross-industry$_standard_process_for_data_mining oldid$ = 1242296671

[6] "Yahoo Finance - Stock Market Live, Quotes, Business & Finance News." [Online]. Available: https://finance.yahoo.com/?guccounter=1guce_referrer = aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8guce_referrer_sig = AQAAAHChNZHxwypD7B3fLCz2hgEDvnm7HQFk5S10W4_GGhPR2Ad75RkG1U4Ue3SpTh01tmL3Ce−vU_O4mkUKj1fGD2jJtw8En1pPzyJpjdx6R7UTBN84a4ZC5Ed−Z8S85TZFqgXqz3SiKJuQ5tfmjkmT1PFIIehNEvjR5M1ySvUwrq

[7] R. Wirth and J. Hipp, "CRISP-DM: Towards a Standard Process Model for Data Mining."